

UDHAYA ADHITHYA VARMAN

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Summary

Passionate artificial intelligence student with a strong foundation in machine learning, want to leverage AI knowledge and skills to contribute innovative solution to real-world challenges in a dynamic organization.

Education

Sathyabama Institute of Science and Technology of Engineering Bachelor of Computer Science Engineering (CSE) – AI CGPA: 8.0	In 4th Year
Siddharatha Junior College Higher Secondary Education CGPA: 6.57	Mar 2022
Gangothri Vidyalaya School Secondary Education CGPA: 9.3	Mar 2018

Projects

- Domain-Specific AI Research Assistant with Knowledge Graph** *Ollama · Hybrid RAG · ChromaDB · Streamlit* | [Github](#)
I developed a Domain-Specific AI Research Assistant with Knowledge Graph that ingests academic PDFs and answers research queries using a Hybrid RAG pipeline combining ChromaDB dense retrieval and BM25 sparse search fused via Reciprocal Rank Fusion. The system verifies every generated claim for hallucinations using an NLI cross-encoder model and auto-builds an interactive multi-paper knowledge graph to enable graph-based reasoning across entities like models, datasets, and metrics.
- Multi-Agent AI Research Assistant** *LLaMA 3.3-70B · Groq · ChromaDB · Semantic Scholar · FastAPI · Streamlit* | [Github](#)
I developed a Multi-Agent AI Research Assistant that orchestrates 5 specialized AI agents — Manager, Searcher, Summarizer, Critic, and Writer — to automate the academic research process end-to-end. The system searches over 200 million papers from Semantic Scholar using custom relevance scoring combining semantic similarity, citation impact, and recency, then produces structured 7-section literature reviews complete with gap analysis and citation validation.
- SignBridge — Real-Time ISL Translator with Edge Deployment** *MediaPipe · CNN-LSTM · ONNX · OpenCV · pytsx3* | [Github](#)
I developed a real-time Indian Sign Language alphabet recognition system that detects and translates all 26 ISL hand signs using a sliding 30-frame CNN-LSTM temporal model with multi-head attention. The system uses MediaPipe Tasks API for automatic single/two-hand landmark extraction, achieves 99.74% validation accuracy trained on CPU in 22 minutes, and deploys via ONNX FP32 at 353 FPS.

Skills

- Languages & Tools:** Python, Streamlit
- Specialized Area:** Artificial Intelligence, Machine Learning, Data Analysis

Certifications

- Certification in Artificial Intelligence
- Certification in Deep Learning Fundamental
- Certification in Machine Learning
- Certification in Prompt Engineering for Everyone
- Certification in Python for Database